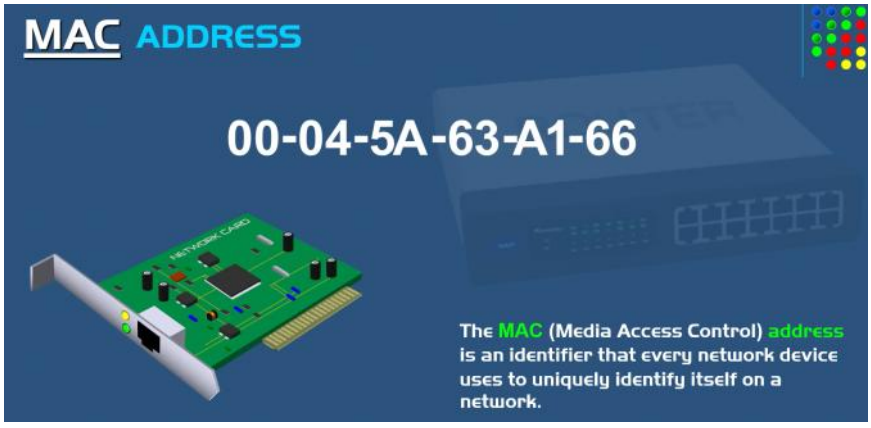
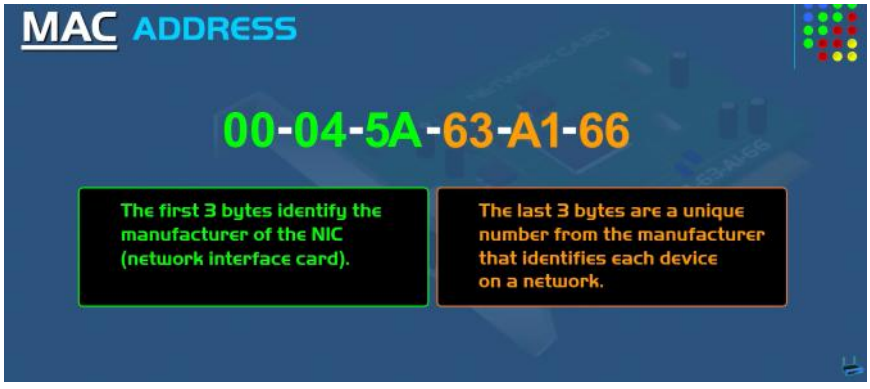
MAC Address

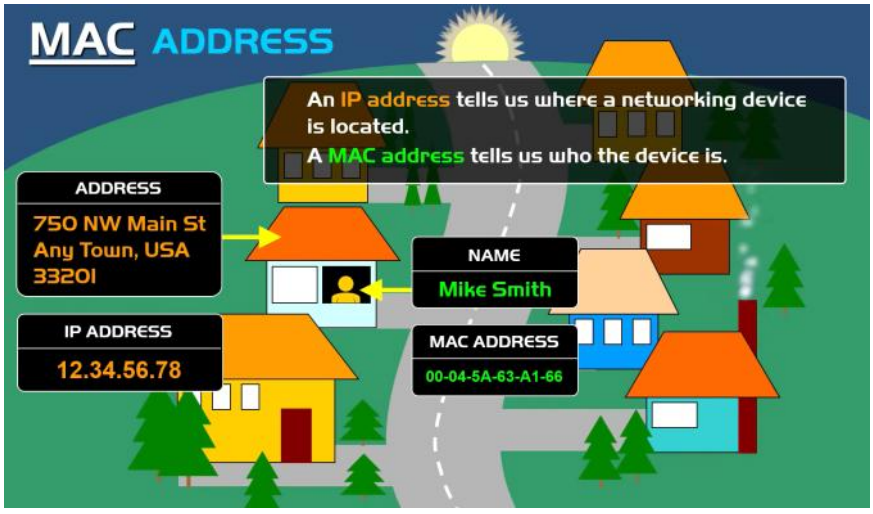
Friday, February 27, 2026 9:12 PM

A MAC (Media Access Control) address is a unique identifier permanently assigned to every network device by its manufacturer. It's a six-byte hexadecimal number that identifies the manufacturer in the first three bytes and the specific device in the last three. Also known as a physical or hardware address, MAC addresses are formatted differently across operating systems (dashes for Windows, colons for Apple/Linux, periods for Cisco).

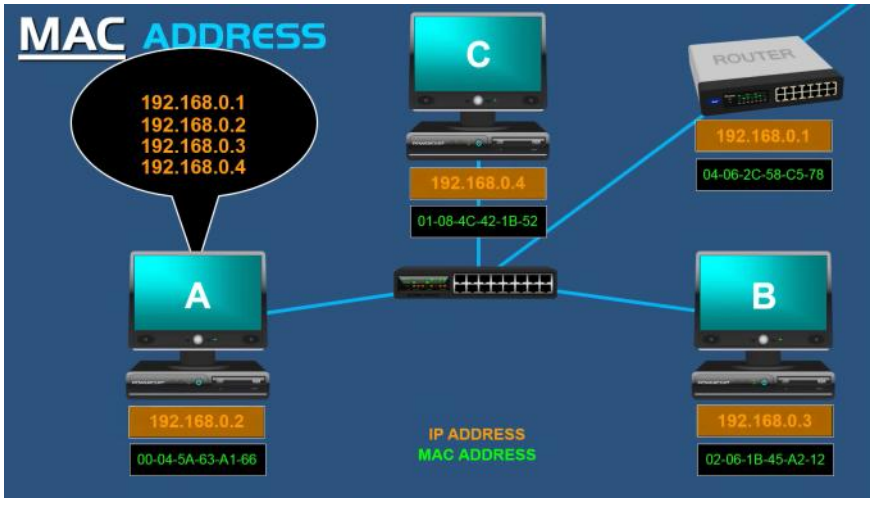


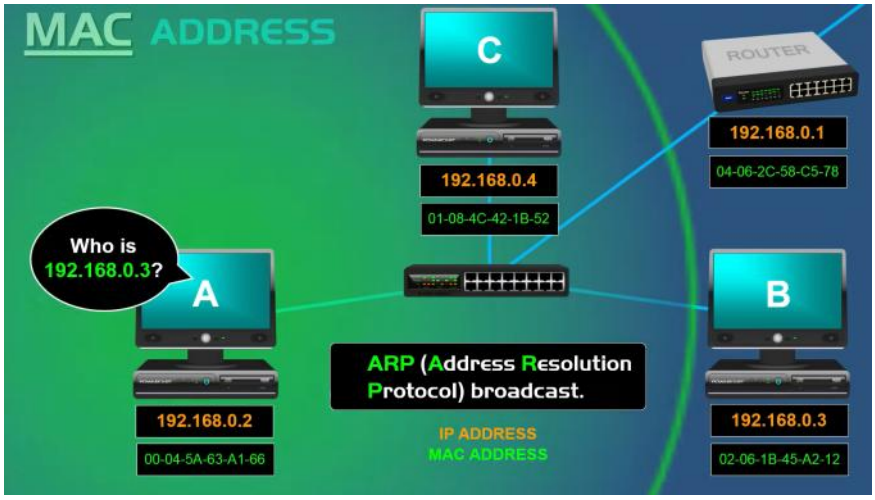


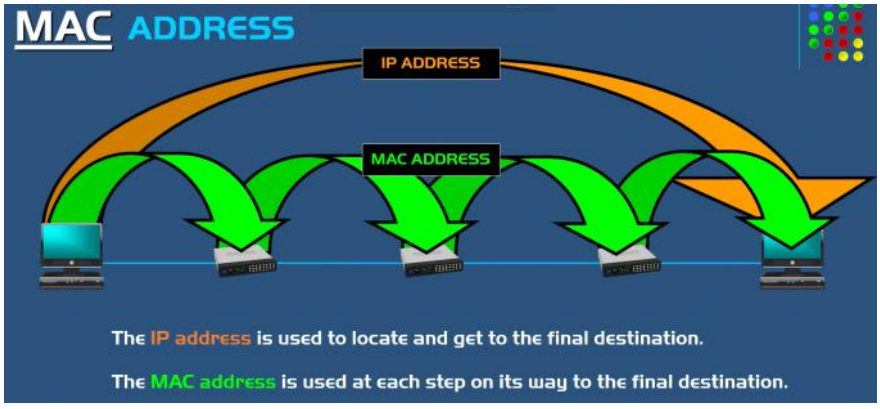
The primary purpose of a MAC address is to enable communication between network devices. While IP addresses are used to locate a device on a network (like a mailing address for a house), MAC addresses specifically identify the device itself (like the name of a person living in the house). IP addresses can change, but MAC addresses are permanent.



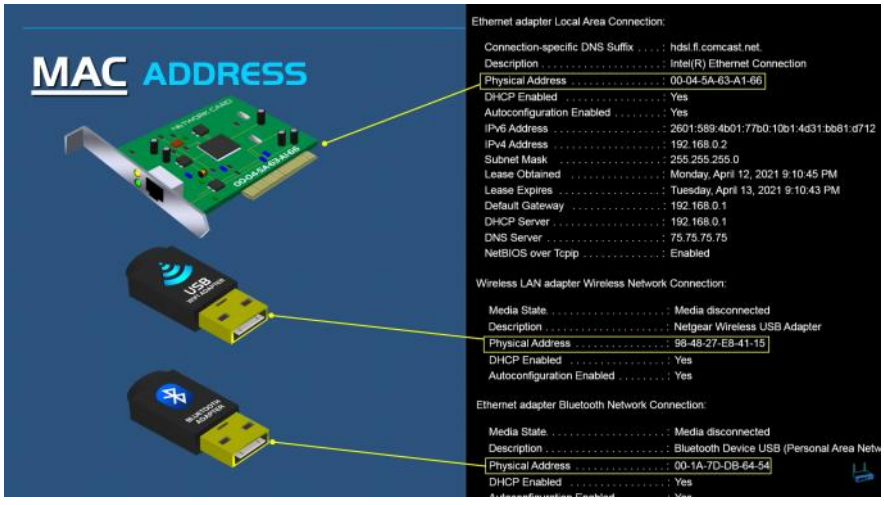
In networking, both IP and MAC addresses work together. When devices on the same local network communicate, an Address Resolution Protocol (ARP) broadcast is used to find the MAC address of the target device. When devices on different networks communicate, data is forwarded through routers, and each router along the path uses MAC addresses to pass the data to the next router until it reaches the final destination. You can find a computer's MAC address using commands like ipconfig /all on Windows or ifconfig on Linux/Mac. A computer can have multiple MAC addresses if it has multiple network interfaces (e.g., wired, wireless, Bluetooth).




MAC ADDRESS
C
192.168.0.4
01-08-4C-42-1B-52
ROUTER
192.168.0.1
04-06-2C-58-C5-78
Who is 192.168.0.3?
A
192.168.0.2
00-04-5A-63-A1-66
ARP (Address Resolution Protocol) broadcast.
IP ADDRESS
MAC ADDRESS
B
192.168.0.3
02-06-1B-45-A2-12


MAC ADDRESS
IP ADDRESS
MAC ADDRESS
The IP address is used to locate and get to the final destination.
The MAC address is used at each step on its way to the final destination.

MAC ADDRESS
To find the MAC address on your computer.
Windows – ipconfig /all
Linux & MAC – ifconfig
Command Prompt
Microsoft Windows
C:\Users\Admin> ipconfig /all
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . . . . : hdsl.fl.comcast.net.
Description . . . . . : Intel(R) Ethernet Connection
Physical Address . . . . . : 00-04-5A-63-A1-66
DHCP Enabled . . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
IPv6 Address . . . . . : 2601:589:4b01:77b0:10b1:4d31:bb81:d712
IPv4 Address . . . . . : 192.168.0.2
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained . . . . . : Monday, April 12, 2021 9:10:45 PM
Lease Expires . . . . . : Tuesday, April 13, 2021 9:10:43 PM
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DNS Server . . . . . : 75.75.75.75
NetBIOS over Tcpip . . . . . : Enabled
Wireless LAN adapter Wireless Network Connection:
Media State. . . . . : Media disconnected
Description . . . . . : Netgear Wireless USB Adapter
Physical Address . . . . . : 98-48-27-E8-41-15


MAC ADDRESS
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IPv6 Address . . . . . : 2601:589:4b01:77b0:10b1:4d31:bb81:d712
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DNS Server . . . . . : 75.75.75.75
NetBIOS over Tcpip . . . . . : Enabled
Wireless LAN adapter Wireless Network Connection:
Media State. . . . . : Media disconnected
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Physical Address . . . . . : 98-48-27-E8-41-15
DHCP Enabled . . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
Ethernet adapter Bluetooth Network Connection:
Media State. . . . . : Media disconnected
Description . . . . . : Bluetooth Device USB (Personal Area Netw
Physical Address . . . . . : 00-1A-7D-DB-64-54
DHCP Enabled . . . . . : Yes